SRM UNIVERSITY–AP, AMARAVATI
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

MACHINE LEARNING CASE STUDY REPORT

on

"TRAFFIC CONGESTION PREDICTION USING MACHINE LEARNING"

A Classification-Based Case Study on the Bangalore City Traffic Dataset

Submitted in partial fulfillment of the requirements for the course
MACHINE LEARNING

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Abstract

Traffic congestion increases travel time, fuel consumption and environmental pollution. This case study applies supervised machine learning to classify each recorded road observation in Bangalore as free-flow, moderate or congested. The Bangalore City Traffic Dataset, obtained from Kaggle, contains 8,936 records and 16 original columns covering traffic volume, average speed, incidents, weather, roadwork and date information for 8 areas and 16 roads between January 2022 and August 2024. The numeric congestion level was converted into three classes using thresholds computed from the training data only. After preprocessing and time-based feature extraction, three classifiers (Decision Tree, Support Vector Machine and K-Nearest Neighbours) were trained on a chronological 70:10:20 split. The SVM achieved the best test performance among the three, with an accuracy of 90.88% and a macro F1-score of 90.04%.

1. Introduction

Traffic conditions differ across roads and locations and change with demand, road capacity, signals, parking, weather, incidents and roadwork. Understanding these conditions helps identify whether a recorded road situation is free-flow, moderate or congested. Machine learning models can learn the relationship between traffic, road, location and environmental information and the resulting congestion condition. Bangalore is widely known for heavy congestion, which makes it a relevant setting for such a study.

The case study question is whether traffic and road information can classify each recorded observation as free-flow, moderate or congested. The objective is to compare three selected classifiers for this task. This is a classification study of recorded observations, not a live or next-hour forecasting service.

2. Dataset Description

The study uses the Bangalore City Traffic Dataset (Kaggle, author: preethamgouda), which is derived from open traffic data of the Government of Karnataka. It contains 8,936 records and 16 columns covering 8 areas and 16 roads or intersections, with observations dated from January 2022 to August 2024. The columns are: date, area name, road or intersection name, traffic volume, average speed, travel time index, congestion level, road capacity utilization, incident reports, environmental impact, public transport usage, traffic signal compliance, parking usage, pedestrian and cyclist count, weather conditions, and roadwork and construction activity. No missing values or duplicate rows were found. The target variable is derived from the numeric congestion level. Table 1 summarises the main numeric variables.

Table 1. Summary statistics of key variables

Variable	Mean	Std. dev.	Min	Max	Missing
Traffic volume	29,236.05	13,001.81	4,233.00	72,039.00	0
Average speed	39.45	10.71	20.00	89.79	0
Congestion level	80.82	23.53	5.16	100.00	0

3. Methodology

The implementation was carried out in Python using Google Colab, Pandas, NumPy, Matplotlib, Seaborn and Scikit-learn. The workflow included dataset loading, chronological sorting, feature preparation, class generation, model training and evaluation, as shown in Figure 1.

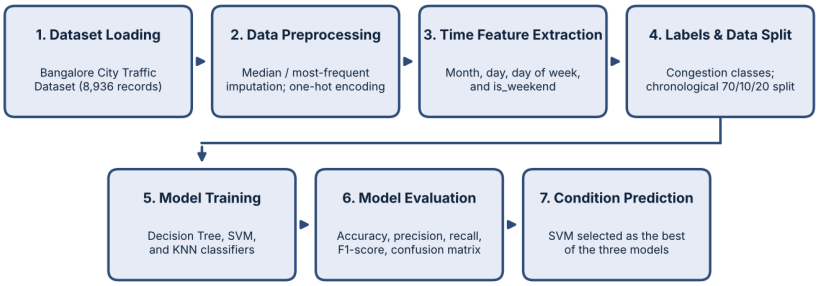


Figure 1. Workflow of Traffic Congestion Prediction Using Machine Learning.

3.1 Data Preprocessing and Features

- The date column was parsed and the 8,936 observations were sorted chronologically. Month, day, day of week and a weekend indicator were extracted from the date.
- Sixteen predictors were used: eight numeric traffic and environmental indicators, four categorical inputs (area, road, weather and roadwork) and the four time features. The date, the target and the proxy columns (travel time index and road capacity utilization) were excluded.
- Missing values would be handled by median imputation (numeric) and most-frequent imputation (categorical); numeric features were standardised and categorical features one-hot encoded. These steps were fitted on the training data only, inside each model pipeline.

3.2 Class Generation and Data Split

The dataset provides a numeric congestion level, which was converted into three classes using cut-offs calculated from the training data only: the 33rd percentile (74.59) and the 66th percentile (100.00). Scores below 74.59 are free-flow, scores from 74.59 up to but not including 100 are moderate, and a score of 100 is congested. Because the congestion level is capped at 100, the congested class consists of records at this maximum, which explains the strong peak at 100 in Figure 2. The data was split chronologically into 6,255 training, 893 validation and 1,788 test records. The test set contained 573 free-flow, 418 moderate and 797 congested records.

3.3 Classification Algorithms and Metrics

Three classifiers were implemented, each in a pipeline with the same preprocessing: a Support Vector Machine (RBF kernel, C = 1, balanced class weights), a Decision Tree (maximum depth 8, balanced class weights) and K-Nearest Neighbours (k = 15). The SVM learns maximum-margin boundaries between the classes, the Decision Tree applies a sequence of simple feature-based questions, and KNN assigns the most common class among the 15 most similar records. The models were compared on validation and test data using accuracy and macro-averaged precision, recall and F1-score, which give equal weight to the three classes, together with the confusion matrix.

4. Implementation and Visualization

The exploratory analysis is shown in Figure 2. The congestion scores are concentrated at the high end, with a strong peak at 100 (mean 80.82, median 92.39). Koramangala has the highest area average (94.0) and Electronic City the lowest (54.5). Monthly average congestion stays broadly steady, between about 77 and 84.

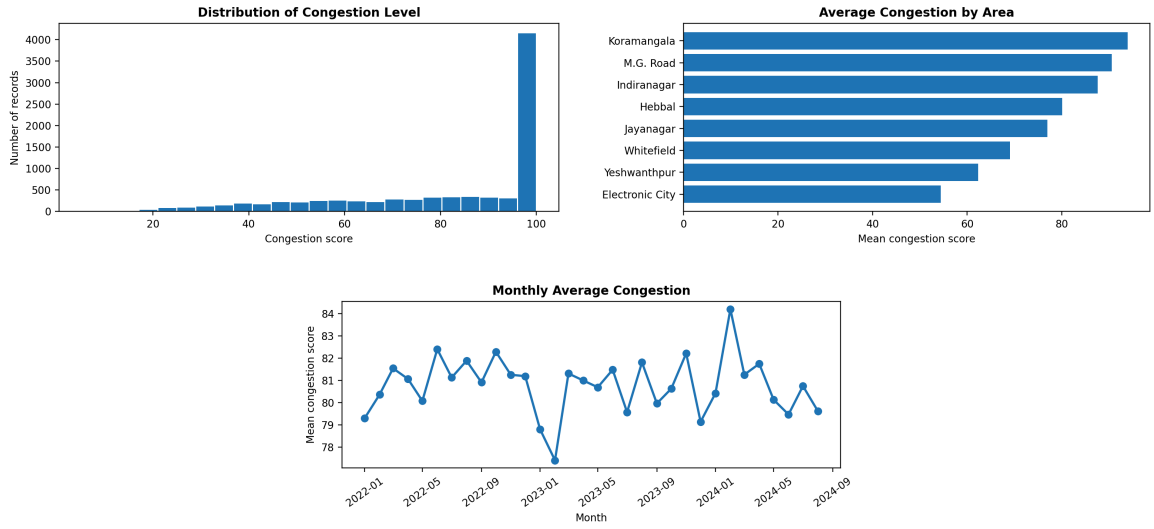


Figure 2. Congestion score distribution, average congestion by area, and monthly average congestion.

5. Results and Discussion

The following values were obtained on the chronological validation and test sets. The SVM achieved the highest test accuracy and F1-score among the three models.

Table 2. Validation and test performance (precision, recall and F1-score are macro averages)

Model	Val. Acc. (%)	Val. F1 (%)	Test Acc. (%)	Test Prec. (%)	Test Recall (%)	Test F1 (%)
Decision Tree	91.71	90.93	90.16	88.81	89.93	89.12
SVM	**91.27**	**90.32**	**90.88**	**89.69**	**91.14**	**90.04**
KNN	82.98	80.23	84.96	82.78	82.83	82.80

The SVM reached 90.88% test accuracy and a 90.04% macro F1-score, slightly ahead of the Decision Tree (90.16%, 89.12%), while KNN performed clearly worse (84.96%, 82.80%), consistent with its sensitivity to feature scaling and to the many one-hot encoded categorical features. The SVM and Decision Tree differ by less than one percentage point in accuracy, and on the validation set the Decision Tree scored slightly higher, so the SVM leads by a narrow margin.

5.1 Class-wise Performance of the SVM

Table 3. Confusion matrix and class-wise metrics of the SVM on the test set

Actual class	Pred. Free-flow	Pred. Moderate	Pred. Congested	Precision	Recall	F1-score	Support
Free-flow	527	46	0	0.96	0.92	0.94	573
Moderate	25	384	9	0.75	0.92	0.82	418
Congested	0	83	714	0.99	0.90	0.94	797

The SVM correctly classified 1,625 of the 1,788 test records (Figure 3). Free-flow and congested records are almost never confused with each other: no free-flow record was predicted as congested and no congested record as free-flow. The main weakness is the moderate class, which has a precision of only 0.75 because 83 congested and 46 free-flow records were predicted as moderate, even though 92% of actual moderate records were found.

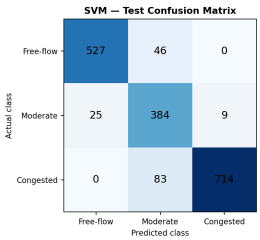


Figure 3. Confusion matrix of the SVM on the test set (rows = actual, columns = predicted).

6. Conclusion and Future Work

This case study analysed 8,936 records across 8 areas and 16 roads and labelled congestion scores using training-only 33rd and 66th percentile cut-offs. Three classifiers were trained to classify each record, and on the 1,788 held-out records the SVM performed best, with 90.88% accuracy and a 90.04% macro F1-score. The results show that traffic, road, location and weather information can classify congestion condition well, although the moderate class remains the hardest to recognise.

The study classifies each record from its own features and does not forecast future traffic. The labels depend on the chosen cut-offs, and the congested class corresponds to the capped value of 100. Future work can connect real-time sensor, GPS, incident, weather and roadwork data; improve minority-class prediction through tuning, class weighting or resampling; explore time-series and graph-based models for rush-hour patterns and congestion spread between connected roads; and build a live dashboard with confidence-aware alerts tested on unseen roads and dates.

References

[1]Bangalore City Traffic Dataset, Kaggle. https://www.kaggle.com/datasets/preethamgouda/banglore-city-traffic-dataset

[2]C. Cortes and V. Vapnik, “Support-vector networks,” Machine Learning, vol. 20, pp. 273–297, 1995.

[3]L. Breiman, J. Friedman, R. Olshen and C. Stone, Classification and Regression Trees. Wadsworth, 1984.

[4]T. Cover and P. Hart, “Nearest neighbor pattern classification,” IEEE Trans. Information Theory, vol. 13, no. 1, pp. 21–27, 1967.

[5]F. Pedregosa et al., “Scikit-learn: Machine Learning in Python,” Journal of Machine Learning Research, vol. 12, pp. 2825–2830, 2011.